measures dissimilarity between probability distributions

but not always satisfied

Formal Setup

- Let α and β be probability measures on $\mathscr{X}$.

- Statistical divergence: a function $D : \mathscr{P}(X) \times \mathscr{P}(X) \rightarrow [0,\infty]$ that **measures dissimilarity between probability distributions**. It satisfies:

  1. **Nonnegativity**: $D(\alpha \,||\, \beta) \geq 0$ for all α, β

  2. **Identity of indiscernibles**: $D(\alpha \,||\, \beta) = 0 \iff \alpha = \beta$

- Nice to have ***but not always satisfied***:

  3. **Symmetry**: $D(\alpha \,||\, \beta) = D(\beta \,||\, \alpha)$

  4. **Triangle Inequality**: $D(\alpha \,||\, \beta) \leq D(\alpha \,||\, \gamma) + D(\beta \,||\, \gamma)$

Part I: f-Divergences